Laboratory/Tutorial Manual with Instructions

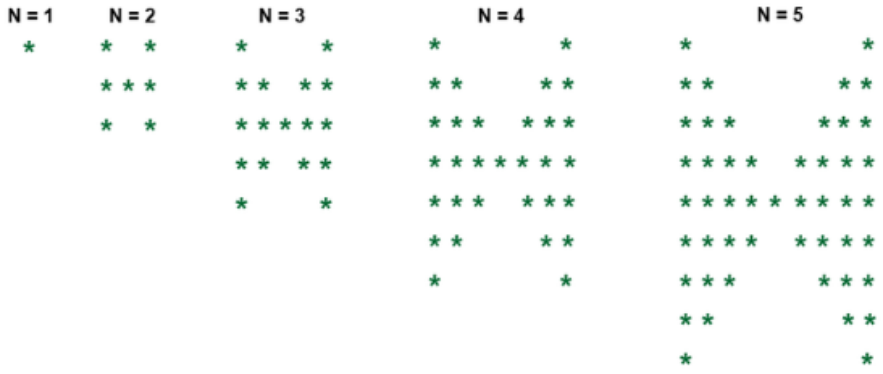
Academic Year	2026-27	**Term**	Odd
Institute	2026-27	**Semester**	1
Department	DICA	**Course Code**	2605CS102
Program	CSE	**Course Name**	Object Oriented Programming using Java
Staff Name	Swati Sharma		

Lab	Category	No.	Programs	Hrs.
Practical -1: C Programs				
1	A	1	Write a program to find a Factor of a given number.[A]	2
	A	2	Write a program to find a sum of numbers entered by user. (E.g: 582=5+8+2=15) [A]	
	A	3	Write a program to find a sum of even number 1D array. [A]	
	B	4	Write a program to find a total odd and total even digit of a given number.[B]	
	C	5	Write a program to find factorial of given number.[C]	
Practical -2: Pattern Programs				

2	A	1	Print the following Patterns: [A]
			`*  *  *  *****  *`
			`**  **  **  ****  **`
			`***  ***  ***  ***  ***`
			`****  ****  ****  **  ****`
			`*****  *****  *****  *  ***`
			`**`
			`*`
				2

	B	2	Print the following Patterns: [B]
			`1  1  d  *****  * * * * *`
			`12  22  da  ####  *  *`
			`123  333  dar  ***  *  *`
			`1234  4444  dars  ##  *  *`
			`12345  55555  darsh  *  * * * * *`
			`darsha`
			`darshan`	

	C	3	**Butterfly Pattern** The butterfly pattern is a **symmetric star (*) pattern** that looks like the shape of a butterfly. It is usually printed using **two mirrored triangles**.	

			
Practical -3: Basic Java Programs			
3	A	1	JDK installation and configuration.[A]
	A	2	Write a java program to print Hello World.[A]
	A	3	Write a java program to take user input [A] i. Through Command Line Argument. ii. Through Scanner class.
	B	4	Write a program to get 2 numbers from the user and print the sum of two numbers using command line and Scanner class.[B]
Practical -4: Programs using operators and data types			
4	A	1	Write a program that reads a number in meters, converts it to feet, and displays the result.[A]
	A	2	Develop a Java program to generate the Fibonacci sequence up to a specified number of terms using recursion.[A]
	B	3	WAP to find weather given number is Armstrong number is not.[B]
	C	4	Write a program that prompts the user to enter a letter and check whether a letter is a vowel or a constant.[C]
Practical -5: Programs using operators and data types			
5	A	1	Illustrate the Operator precedence.[A] a. 10 + 20 * 30 b. 100 + 200 ÷ 10 − 3 × 10 c. 50 × 4 + 80 ÷ 8 − 5 d. 100 + 200 / 10 – 3 * 10 e. 400 − 50 × 3 + 90 ÷ 9 f. 250 + 100 ÷ 20 × 6 – 30 g. 250 − 100 ÷ (5 + 5) + 12 × 3
	A	2	Write a program to create basic calculator by getting 2 numbers and an operation(+,-,/,*,%) from the user and apply the operation given in a string on the given numbers.[A]
	A	3	Write a program to calculate the area of Rectangle.[A]
	B	4	Write a program to convert temperature from Fahrenheit to Celsius. (Formula : c = f-32*5/9) [B]
	C	5	Write a program that prompts the user to enter three numbers. Find the largest number.[C]

Practical -6: Programs using operators and data types				
6	A	1	A student gets scholarship if: Percentage ≥ 75 OR Annual income < ₹2,50,000. Write a java program to retrieve 3 subject marks, find percentage and then evaluate particular student is eligible to scholarship or not.[A]	2
	A	2	Write a java program to find that entered no is odd or even without using arithmetic operator. [Hint: Bitwise operator][A]	
	B	3	A power company charges:[B] ₹5 per unit for first 100 units ₹7 per unit for next 100 units ₹10 per unit beyond 200 units Write a Java program to calculate the total electricity bill using arithmetic operators.	
	C	4	Write a java program to compute shopping bill[C] Discount rules: • Bill ≥ ₹5000 → 20% discount • Bill ≥ ₹3000 → 10% discount Add 5%GST to final bill amount. Calculate final payable amount.	
Practical -7: Programs using if-else ladders, conditional and branching statement				
7	A	1	The marks obtained by a student in 5 different subjects are input through the keyboard. The student gets a division as per the following rules: I. Percentage above or equals to 60-first division II. Percentage between 50 to 59-second division III. Percentage between 40 and 49-Third division IV. Percentage less than 40-fail Write a program to calculate the division obtained by the student.[A]	2
	A	2	Write a program to check whether a number is even or odd.[A]	
	A	3	Write a program to find maximum no from given 3 no.(without if-else).[A]	
	B	4	Write a Java program to simulate an ATM system where invalid transactions are skipped using continue and the program exits on user request using break.[B]	
	C	5	Write a Java program to implement a menu-driven area calculator for Circle, Rectangle, Triangle, and Square using a switch statement.[C]	
Practical -8: Programs using if-else ladders, conditional and branching statement				
8	A	1	Write a program to check that the given number is prime or not.[A]	2
	B	2	Write a program to check whether a year is leap year or not.[B]	
	B	3	Write a program to find that given Number is palindrome or not. [B]	
	C	4	Write a program in Java to print all 3-digit prime numbers.[C]	
	C	5	Write a Java program that accepts a Roman numeral (I, V, X, L, C) and displays its decimal value using a switch statement.	
Practical -9: Programs using Array				
9	A	1	Write a Java program to sum values of an array.[A]	2
	A	2	Write a program that creates and initializes a four integer element array. Calculate and display the average of its values.[A]	
	A	3	Write a program to print given array in reverse order.[A]	
	B	4	Write a Java program to copy all the elements of an array to another array and print both the array elements. [B]	

	B	5	Implement an array with 5 elementS. Create four methods for array operations (sortArray(), searchArray(), sumArray(), and avgArray()) and call all the four methods using object.[B]	
	C	6	Dominant Number in an Array. A dominant number is a number whose frequency of occurrence in an array is strictly greater than the count of unique elements present in that array. You are given an integer array.[C] Your task is to identify and return all dominant numbers from the array. Input: arr = [1, 2, 3, 1, 1, 4, 5, 1, 1, 1] Output: 1 Input: arr = [1, 1, 1, 2, 2, 2, 3] Output: 1, 2	
Practical -10: Programs using Advance Array and String				
10	A	1	Write a program to find length of string and print second half of the string.[C]	
	A	2	Write an interactive program to print a string entered in a pyramid form. For instance, the string "stream" has to be displayed as follows:[B] s st str stre strea stream	2
	A	3	Find Maximum, Second Maximum, Minimum, and Second Minimum from an Array This program takes an array of integers as input and identifies: 1. the Maximum element, 2. the Second Maximum element, 3. the Minimum element, and 4. the Second Minimum element. The array must contain at least four distinct elements to determine all four values correctly. Input: Array = [10, 5, 20, 8, 15] Output: Maximum = 20 Second Maximum = 15 Minimum = 5 Second Minimum = 8 Input: Array = [3, 7, 1, 9, 4] Output: Maximum = 9 Second Maximum = 7 Minimum = 1 Second Minimum = 3	

	B	4	Simqual String Problem Two strings are called Simqual strings if they satisfy the following conditions: 1. Both strings must have the same length 2. Both strings must contain the same characters 3. The frequency of each character must be equal in both strings 4. Order of characters does not matter The program should check whether the given two strings are Simqual or Not Simqual. Input: String 1 = "listen" String 2 = "silent" Output: Strings are Simqual Input: String 1 = "hello" String 2 = "world" Output: Strings are NOT Simqual	
	C	5	There is an integer array nums sorted in ascending order (with distinct values). Prior to being passed to your function, nums is possibly rotated at an unknown pivot index k (1 <= k < nums.length) such that the resulting array is [nums[k], nums[k+1], ..., nums[n-1], nums[0], nums[1], ..., nums[k-1]] (0-indexed). For example, [0,1,2,4,5,6,7] might be rotated at pivot index 3 and become [4,5,6,7,0,1,2]. Given the array nums after the possible rotation and an integer target, return the index of target if it is in nums, or -1 if it is not in nums.[C] Example 1: Input: nums = [4,5,6,7,0,1,2], target = 0 Output: 4 Example 2: Input: nums = [4,5,6,7,0,1,2], target = 3 Output: -1	
Practical -11: Programs using Class and Object				
11	A	1	Create class Student with attributes (name: String, roll_no:int, SPI:double, course: String). Implement getter() and setter() method to assign data for 3 students and display it.[A]	2
	B	2	Create class Cube with attributes (height, width and depth with double datatype). Implement getter() and setter() method to assign data for 2 cubes. Create method volume() to calculate volume for 2 cubes. [A]	
	C	3	Create class BankDemo and Account(accNum:int, accType:String, balance: double). Implement getter() and setter() method to assign and display data for 3 students.[A]	
Practical -12: Programs using Constructor and Overloading				
12	A	1	Develop a program that validates student marks using a constructor and displays whether the student is eligible for placement.	2
	B	2	Write a Java program to create a class called Employee with a name, job title, and salary attributes, and methods to calculate and update salary.	
	C	3	Create a class Book with three overloaded constructors: Default constructor Constructor accepting title Constructor accepting title and price	

Practical -13: Programs using Advanced Object Oriented Concepts				
13	A	1	Write a program to create circle class with area function to find area of circle. [A]	2
	A	2	Write a Java program to find the angle between the hour and minute hands.[A]	
	B	3	Create a class which ask the user to enter a sentence, and it should display count of each vowel type in the sentence. The program should continue till user enters a word “quit”. Display the total count of each vowel for all sentences. [B]	
	C	4	Define class for Complex number with real and imaginary as data members. Create its constructor, overload the constructors. Also define addition method to add two complex objects. [C]	
Practical -14: Programs using Static Keyword & Array of Objects				
14	A	1	Write a program to add two different Time class’s objects. Time class contains 3 fields hours, Minute and Seconds. (Take input from user and assume user will always enter a valid data, e.g. if Time1 is 02:55:55 and Time2 is 03:07:10 then result is 06:03:05) [A]	2
	A	2	Write a java program that counts the number of objects created using static. [A]	
	B	3	Write a Java program to create a class called BankAccount with instance variables 'accountNo' and balance, and static variables bankName and interestRate. Implement static methods to get and set the static variables. Create three BankAccount objects and print their details along with the static variables. [B]	
	C	4	Create a Time class to represent time in hours, minutes, and seconds.[C] • Initialize Hr : Min : Sec using a constructor. • Create an AddSeconds(seconds) method to add the given seconds to the initialized time. • Create a SubtractSeconds(seconds) method to subtract the given seconds from the initial time. • Ensure proper time adjustment (60 seconds = 1 minute, 60 minutes = 1 hour). • Print the time in Hr:Min:Sec format. **Input**: Intial Time = 02 : 30 : 45 Add Seconds = 50 Subtract Seconds = 30 Output: Time after adding seconds = 02 : 31 : 35 Time after subtracting seconds = 02 : 31 : 05	
Practical -15: Programs using Static Keyword & Array of Objects				
15	A	1	Write a Java program to create a class called Area with a static final variable PI=3.14159. Implement a method to calculate the area of a circle given its radius.[A]	2
	A	2	Write a Java program to store employee details using instance variables and maintain a common company name using a static variable. Display employee information to illustrate memory sharing through static members.[A]	
	B	3	Write a Java program to create a class called House with private instance variables address, numberOfRooms, and area. Provide public getter and setter methods to access and modify these variables. Add a method called calculatePrice() that returns the price of the house based on its area and a price per square meter.[B]	
	C	4	Design a program to compare two student objects and return the object having higher marks.[C]	

Practical -16: Programs using inheritance				
16	A	1	Declare a class called Student having following data members:id_no, no_of_subjects_registered, subject_code, subject_credits, grade_obtained and spi. Define constructor and calculate_spi methods. Define main to instantiate an array for objects of class student to process data of n students [A]	2
	A	2	Write a Java program to create a class called Employee with methods called work() and getSalary(). Create a subclass called HRManager that overrides the work() method and adds a new method called addEmployee().[A]	
	B	3	Write a Java program for a University Management System where the Student and Faculty classes inherit common attributes and methods from a parent Person class. Demonstrate code reusability through inheritance.[B]	
	C	4	Develop a Java program for a **Banking System** where `Account` is inherited by `SavingsAccount`, and `SavingsAccount` is further inherited by `LoanAccount`. Display account and loan details using multilevel inheritance.[C]	
Practical -17: Advanced Programs using inheritance				
17	A	1	Demonstrate the use of Super Keyword. [A]	
	A	2	Write a Java program for a **Payment Processing System** where different payment methods such as Credit Card, UPI, and Net Banking override a common `processPayment()` method and are invoked using parent class references.[A]	
	A	3	Declare a class called book having author_name as private data member. Extend book class to have two sub classes called book_publication & paper_publication. Each of these classes have private member called title. Write a complete program to show usage of dynamic method dispatch (dynamic polymorphism) to display book or paper publications of given author. [A]	
	B	4	Develop a Java-based Hospital Management System where a parent class Hospital contains a method to display general hospital information, and a derived class Department overrides this method to display department-specific details. The derived class should use the super keyword to invoke the overridden method of the parent class before displaying its own information. The program should demonstrate inheritance, method overriding, and the use of the super keyword to access parent class functionality within a child class.[B]	
	C	5	Design a class named MyPoint to represent a point with x- and y-coordinates. The class contains: The data fields x and y that represent the coordinates with getter methods. 1. a no-arg constructor that creates a point (0, 0). 2. a constructor that constructs a point with specified coordinates. 3. a method named distance that returns the distance from this point to a specified point of the MyPoint type. 4. a method named distance that returns the distance from this point to another point with specified x- and y-coordinates.Create a class named ThreeDPoint to model a point in a three-dimensional space. Let ThreeDPoint be derived from MyPoint with following additional features: 1. A data fields named z that represents the z-coordinate. 2. A no-arg constructor that creates a point (0, 0, 0). 3. A constructor that constructs a point with three specified coordinates. 4. A get method that returns the z value. 5. Override the distance method to return the distance between two points in the three-dimensional space. Write a program that creates two points (0, 0, 0) and (10, 30, 25.5) and display the distance between the two points. [C]	

Practical 18-: Programs using Abstract class				
18	A	1	The abstract vegetable class has three subclasses named Potato, Brinjal and Tomato. Write a java program that demonstrates how to establish this class hierarchy. Declare one instance variable of type String that indicates the color of a vegetable. Create and display instances of these objects. Override the toString() method of object to return a string with the name of vegetable and its color. [A]	2
	A	2	Write a Java program to create an abstract class BankAccount with abstract methods deposit() and withdraw(). Create subclasses: SavingsAccount and CurrentAccount that extend the BankAccount class and implement the respective methods to handle deposits and withdrawals for each account type.[A]	
	B	3	Write a Java program to create an abstract class Vehicle with abstract methods startEngine() and stopEngine(). Create subclasses Car and Motorcycle that extend the Vehicle class and implement the respective methods to start and stop the engines for each vehicle type.[B]	
Practical -19:Programs using Dynamic method Dispatch				
19	A	1	Write a Java program for a Vehicle Management System where a parent class Vehicle is inherited by classes such as Car, Bike, and Bus. Use a parent class reference to invoke overridden methods at runtime and demonstrate Dynamic Method Dispatch for displaying vehicle-specific information.[A]	2
	B	2	Write a Java program for a Hospital Management System where different departments such as Cardiology, Neurology, and Orthopedics override a common displayServices() method. Use a parent class reference to dynamically call department-specific methods at runtime.[B]	
	B	3	Develop a Java program for a Food Delivery System where different restaurant categories such as FastFood, SouthIndian, and ChineseRestaurant override a common prepareOrder() method. Use a parent class reference to dispatch method calls dynamically at runtime.[B]	
	C	4	Write a Java program for a Library Management System where different membership types such as StudentMember, FacultyMember, and ResearchMember override a method to determine borrowing limits. Demonstrate Dynamic Method Dispatch to enforce membership-specific borrowing policies.[C]	
Practical -20:Programs using Interface				
20	A	1	Write a program that illustrates interface inheritance. Interface A is extended by A1 and A2. Interface A12 inherits from both A1 and A2.Each interface declares one constant and one method. Class B implements A12.Instantiate B and invoke each of its methods. Each method displays one of the constants. [A]	2
	A	2	Create interface EventListener with performEvent() method. Create MouseListener interface which inherits EventListener along with mouseClicked(), mousePressed(), mouseReleased(), mouseMoved(), mouseDragged() methods. Also create KeyListener interface which inherits EventListener along with keyPressed(), keyReleased() methods. WAP to create EventDemo class which implements MouseListener and KeyListener and demonstrate all the methods of the interfaces. [A]	

	B	3	The Transport interface declares a deliver () method. The abstract class Animal is the super class of the Tiger, Camel, Deer and Donkey classes. The Transport interface is implemented by the Camel and Donkey classes. Write a test program that initialize an array of four Animal objects. If the object implements the Transport interface, the deliver () method is invoked. [B]	
Practical -21: Programs using Advanced Interface				
21	A	1	Write a program in Java to create an interface Car with moveForward, moveBackward, moveLeft, moveRight, applyBreak methods, create a class named Swift and Thar which implement the Car interface. [A]	2
	B	2	Write a Java program to create an interface Playable with a method play() that takes no arguments and returns void. Create three classes Football, Volleyball, and Basketball that implement the Playable interface and override the play() method to play the respective sports. [B]	
	C	3	Create an interface Device containing a method powerOn(). Create another interface SmartDevice that extends Device and contains a method connectInternet(). Implement both interfaces in a class SmartPhone. Display the functionality of the smartphone using interface references.[C]	
Practical -22: Programs using Exception handling				
22	A	1	Write a program to demonstrate Arithmetic Exception and ArrayIndexOutOfBounds Exception using try-catch block. [A]	2
	A	2	Write a program to create Account class, which is representing a bank account where we can deposit and withdraw money. If user need to withdraw money which exceed our minimum bank balance then it will not be allowed, and will throw InSufficientFundException(Custom Exception). Handle above exception and display proper error message. [A]	
	C	3	Write a program to create a custom exception named AgeException, get the age of the user from command prompt (terminal) and if age is less than 18 it should throw AgeException, also handle this exception in main method. [A]	
Practical -23: Programs using Advanced Exception handling				
23	A	1	Write a program to create Account class, which is representing a bank account where we can deposit and withdraw money. If user need to withdraw money which exceed our minimum bank balance then it will not be allowed, and will throw InSufficientFundException(Custom Exception). Handle above exception and display proper error message. [A]	2
	B	2	Write a Java program that divides two numbers. If Num1 or Num2 were not an integer, the program would throw a Number Format Exception. If Num2 were Zero, the program would throw an Arithmetic Exception. Display appropriate message for each exception. [B]	
	C	3	Write a program in java if number is less than 10 and greater than 50, it generate the exception out of range. Else it displays the square of number. [C]	

Practical -24: Programs using Multithreading				
24	A	1	Write a java Multithread program [A] i. Using Thread class. ii. Using Runnable Interface.	2
	A	2	Write an application that executes two threads. One thread displays "Good Morning" every 1000 milliseconds & another thread displays "Good Afternoon" every 3000 milliseconds. Create the threads by implementing the Runnable interface. [A]	
	B	3	Write a program to create two threads, one thread will print odd numbers and second thread will print even numbers between 1 to 20 numbers. [A]	
Practical -25: Programs using Packages				
25	A	1	Write a java program i. Create a pakage 'Darshan' with class 'BTech' within it. ii. Create a class 'University' to import package 'Darshan' and call the methods of BTech class.	2
	A	2	Write a Java application for a University Management System where academic-related classes such as Student and Faculty are placed in a package named university. Create another package named administration containing a class that accesses the Student and Faculty classes using the import statement. The program should display student and faculty information, demonstrating how packages help organize related classes and how the import statement allows classes from one package to be used in another package.	
	B	3	Develop a Java application for a Bank Account Management System where account details such as account number, customer name, and balance are protected using different access modifiers. Declare the account balance as private, customer name as protected, account type with default access, and account number as public. Create classes within the same package and in different packages to demonstrate which members can be accessed and which are restricted according to Java's access control rules.	
	C	4	Write a Java application for a **Scientific Calculator** that performs various mathematical operations such as square root, power calculation, and value of π. Use the `static import` statement to directly access static methods and constants from Java's `Math` class without prefixing them with the class name. The program should calculate the area of a circle, the square root of a number, and the power of a number, demonstrating the static imports.	
Practical -26: Programs using Advanced Multithreading				
26	A	1	Write a program to implement the solution to producer consumer problem in Java. [A]	2
	A	2	Write a java program to print multiplication table of two numbers using Thread Synchronization.[A]	
	B	3	Write a java program that implements a multi-thread application that has three threads. First thread generates random integer every 1 second and if the value is even, second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of cube of the number. [B]	
	C	4	Write a complete multi-threaded program to meet following requirements: a. Read matrix [A] m x n b. Create m number of threads c. Each thread computes summation of elements of one row, i.e. ith row of the matrix is processed by i^{th} thread. Where 0 <= i< m. d. Print the results. [C]	

Practical -27: Programs using IO programming				
27	A	1	Write a program to replace all “word1” by “word2” from a file1, and output is written to file2 file and display the no. of replacement. [A]	2
	B	2	Write a java program to copy the content of one file to another using Byte and Character Stream both.[B]	
	C	3	Create a class called Student. Write a student manager program to manipulate the student information from files by using FileInputStream and FileOutputStream. [C]	
Practical -28 : Programs using Advanced IO programming				
28	A	1	Write a program that counts number of characters, words, and lines in a file. Use exceptions to check whether the file that is read exists or not. [A]	2
	B	2	Write an application that reads a file and counts the number of occurrences of digit 5. Supply the file name as a command-line argument. [B]	
	C	3	Write a java program to copy the content of one file to another using:[C] i. Buffered Input/Output Stream ii. Buffered Reader and Buffered Writer	
Practical -29 : Programs using Collection Framework				
29	A	1	Write a java program to insert Employee name using Array List interface.[A]	2
	B	2	Write a java program to insert Product price using LinkedList Interface.[B]	
	C	3	Write a program to demonstrate the use of Array List to store and display List of Student class with StudentID, StudentName, StudentRollNo and StudentSPI. [C]	
	C	4	Write a program to sort as per SPI from the List of students stored as ArrayList in previous program. [C]	
Practical -30 : Java Application				
30	A	1	**Food Ordering System** Definition: Implement a Food Ordering System where food item information is encapsulated within classes. Create an abstract class Food Item and derive classes such as Pizza and Burger. Implement polymorphism by overriding price calculation methods. Use constructors to initialize food details, access specifier for controlled access, and static members to count the total orders placed.	2
	B	2	Write a java program to generate Food Ordering System bill in foodBill.txt file	